

MySQL Data Types

Objectives

- Identify the different data types in MySQL and its usage.

Data Types in MySQL

- ***Data types are referred as the type of data, like string, integer, date-time etc., that can be stored in a column (field) of a table.***
- Properly defining the fields in a table is important to the overall optimization of your database.
- *You should use only the type and size of field you really need to use.*

Data Types in MySQL

- *When defining columns in a database, choose a data type that matches the actual size and type of data you wish to store, rather than unnecessarily allocating more space than needed.*

Data Types in MySQL

In MySQL, data types is categorized into three main groups –

- Numeric
- Date and Time
- String

Data Types in MySQL

- An SQL developer must decide what type of data that will be stored inside each column when creating a table.
- *The data type is a guideline for SQL to understand what type of data is expected inside of each column, and it also identifies how SQL will interact with the stored data.*

Numeric Data Types

Data type	Description
BIT(<i>size</i>)	A bit-value type. The number of bits per value is specified in <i>size</i> . The <i>size</i> parameter can hold a value from 1 to 64. The default value for <i>size</i> is 1.
TINYINT(<i>size</i>)	A very small integer. Signed range is from -128 to 127. Unsigned range is from 0 to 255. The <i>size</i> parameter specifies the maximum display width (which is 255)
BOOL	Zero is considered as false, nonzero values are considered as true.
BOOLEAN	Equal to BOOL
SMALLINT(<i>size</i>)	A small integer. Signed range is from -32768 to 32767. Unsigned range is from 0 to 65535. The <i>size</i> parameter specifies the maximum display width (which is 255)
MEDIUMINT(<i>size</i>)	A medium integer. Signed range is from -8388608 to 8388607. Unsigned range is from 0 to 16777215. The <i>size</i> parameter specifies the maximum display width (which is 255)
INT(<i>size</i>)	A medium integer. Signed range is from -2147483648 to 2147483647. Unsigned range is from 0 to 4294967295. The <i>size</i> parameter specifies the maximum display width (which is 255)
INTEGER(<i>size</i>)	Equal to INT(<i>size</i>)

Numeric Data Types

<code>BIGINT(<i>size</i>)</code>	A large integer. Signed range is from -9223372036854775808 to 9223372036854775807. Unsigned range is from 0 to 18446744073709551615. The <i>size</i> parameter specifies the maximum display width (which is 255)
<code>FLOAT(<i>size</i>, <i>d</i>)</code>	A floating point number. The total number of digits is specified in <i>size</i> . The number of digits after the decimal point is specified in the <i>d</i> parameter. This syntax is deprecated in MySQL 8.0.17, and it will be removed in future MySQL versions
<code>FLOAT(<i>p</i>)</code>	A floating point number. MySQL uses the <i>p</i> value to determine whether to use FLOAT or DOUBLE for the resulting data type. If <i>p</i> is from 0 to 24, the data type becomes FLOAT(). If <i>p</i> is from 25 to 53, the data type becomes DOUBLE()
<code>DOUBLE(<i>size</i>, <i>d</i>)</code>	A normal-size floating point number. The total number of digits is specified in <i>size</i> . The number of digits after the decimal point is specified in the <i>d</i> parameter
<code>DOUBLE PRECISION(<i>size</i>, <i>d</i>)</code>	
<code>DECIMAL(<i>size</i>, <i>d</i>)</code>	An exact fixed-point number. The total number of digits is specified in <i>size</i> . The number of digits after the decimal point is specified in the <i>d</i> parameter. The maximum number for <i>size</i> is 65. The maximum number for <i>d</i> is 30. The default value for <i>size</i> is 10. The default value for <i>d</i> is 0.
<code>DEC(<i>size</i>, <i>d</i>)</code>	Equal to DECIMAL(<i>size</i> , <i>d</i>)

String Data Types

Data type	Description
CHAR(size)	A FIXED length string (can contain letters, numbers, and special characters). The <i>size</i> parameter specifies the column length in characters - can be from 0 to 255. Default is 1
VARCHAR(size)	A VARIABLE length string (can contain letters, numbers, and special characters). The <i>size</i> parameter specifies the maximum column length in characters - can be from 0 to 65535
BINARY(size)	Equal to CHAR(), but stores binary byte strings. The <i>size</i> parameter specifies the column length in bytes. Default is 1
VARBINARY(size)	Equal to VARCHAR(), but stores binary byte strings. The <i>size</i> parameter specifies the maximum column length in bytes.
TINYBLOB	For BLOBs (Binary Large Objects). Max length: 255 bytes
TINYTEXT	Holds a string with a maximum length of 255 characters
TEXT(size)	Holds a string with a maximum length of 65,535 bytes

String Data Types

BLOB(size)	For BLOBs (Binary Large Objects). Holds up to 65,535 bytes of data
MEDIUMTEXT	Holds a string with a maximum length of 16,777,215 characters
MEDIUMBLOB	For BLOBs (Binary Large Objects). Holds up to 16,777,215 bytes of data
LONGTEXT	Holds a string with a maximum length of 4,294,967,295 characters
LOBLOB	For BLOBs (Binary Large Objects). Holds up to 4,294,967,295 bytes of data
ENUM(val1, val2, val3, ...)	A string object that can have only one value, chosen from a list of possible values. You can list up to 65535 values in an ENUM list. If a value is inserted that is not in the list, a blank value will be inserted. The values are sorted in the order you enter them
SET(val1, val2, val3, ...)	A string object that can have 0 or more values, chosen from a list of possible values. You can list up to 64 values in a SET list

Date and Time Data Types

Data type	Description
DATE	A date. Format: YYYY-MM-DD. The supported range is from '1000-01-01' to '9999-12-31'
DATETIME(<i>fsp</i>)	A date and time combination. Format: YYYY-MM-DD hh:mm:ss. The supported range is from '1000-01-01 00:00:00' to '9999-12-31 23:59:59'. Adding DEFAULT and ON UPDATE in the column definition to get automatic initialization and updating to the current date and time
TIMESTAMP(<i>fsp</i>)	A timestamp. TIMESTAMP values are stored as the number of seconds since the Unix epoch ('1970-01-01 00:00:00' UTC). Format: YYYY-MM-DD hh:mm:ss. The supported range is from '1970-01-01 00:00:01' UTC to '2038-01-09 03:14:07' UTC. Automatic initialization and updating to the current date and time can be specified using DEFAULT CURRENT_TIMESTAMP and ON UPDATE CURRENT_TIMESTAMP in the column definition
TIME(<i>fsp</i>)	A time. Format: hh:mm:ss. The supported range is from '-838:59:59' to '838:59:59'
YEAR	A year in four-digit format. Values allowed in four-digit format: 1901 to 2155, and 0000. MySQL 8.0 does not support year in two-digit format.

The ENUM

- The ENUM data type in MySQL is used to store a list of predefined values — it's like saying, "the column can only have one value from this specific set."

```
CREATE TABLE users
(
    id INT AUTO_INCREMENT PRIMARY KEY,
    status ENUM('active', 'inactive', 'pending')
);
```

The ENUM

Here, the status column can **only** have:

- 'active'
- 'inactive'
- 'pending'

If you try to insert something like 'deleted', MySQL will reject it.

The ENUM

Key Points about ENUM:

- It's stored internally as numbers (1, 2, 3...) based on the order of values.
- It saves space compared to using plain VARCHAR.
- You can set a default value if you want.

Example:

```
status ENUM('active', 'inactive', 'pending')  
DEFAULT 'pending'
```

The SET

- The SET data type in MySQL is kind of like ENUM, **but** it allows you to store **multiple values** at once — not just one!
- Use SET when a column can have **zero, one, or several options** from a predefined list.

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Example:

```
CREATE TABLE user_preferences (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    notifications SET('email', 'sms', 'push')  
);
```


The SET

The notifications column can store:

- 'email'
- 'sms'
- 'push'

Any combination like 'email,sms' or 'sms,push' Or even **none** (empty).

Storing audio, images, or multimedia files

- A BLOB is designed to store **binary data** like images, audio, or multimedia files — basically anything that isn't plain text.

```
CREATE TABLE product_images
(
    id INT AUTO_INCREMENT PRIMARY KEY,
    image BLOB
);
```

Storing audio, images, or multimedia files

There are several kinds of BLOBs depending on the size you need:

- TINYBLOB (up to 255 bytes)
- BLOB (up to 65 KB)
- MEDIUMBLOB (up to 16 MB)
- LONGBLOB (up to 4 GB)
- **For typical images**, BLOB or MEDIUMBLOB is usually enough.

MySQL Data Types Reference

- <https://dev.mysql.com/doc/refman/8.4/en/data-types.html>